

London Crime Dashboard

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Methodology & Limitations

Why SARIMA?

	Method	MAE	RMSE
0	ARIMA(1,1,0)	6217.2604	
1	Naive Seasonal	1983.5	

Plain ARIMA failed to capture seasonality; SARIMA(1,1,0)(1,1,0,12) closed most of the gap to a naive seasonal baseline while still capturing genuine turning points.

Ensembling Results

	Method	MAE
0	SARIMA alone	
1	Naive Seasonal alone	
2	Ensemble (0.4/0.6 blend)	

SARIMA alone was the weakest method in the original March–June backtest and validation windows. A follow-up test using July 2026's real released data found the Ensemble (0.35% error) outperformed both SARIMA alone (1.20%) and Naive Seasonal alone (1.38%) — across two independent real-world windows, the Ensemble was never the worst-performing method. Because of this, Ensemble forecasts for Aug–Dec 2026 are shown alongside SARIMA throughout this dashboard, rather than replacing it, since no real outcome data yet exists to confirm which method will prove more accurate for those specific months.

Known Limitations

- Public Order Offences:** a real structural drop in recorded incidents began March 2026, likely due to an MPS recording/classification change. The model doesn't know about this, so its forecast for this category (London-wide and per-borough) is likely too high. Excluded from rising/falling rankings but shown in full breakdowns with this caveat.
- Westminster volatility:** highest crime volume and highest volatility (CV = 2.03) of any borough — largely tourism/footfall driven rather than deprivation-driven (confirmed via IMD correlation). Forecasts for Westminster carry more uncertainty, and independent model refits have shown meaningfully different error rates for this borough.
- Small-borough effect:** very small boroughs (e.g. Richmond upon Thames) show higher percentage error due to low absolute crime counts — small numbers amplify percentage swings.
- Model refit sensitivity:** re-fitting SARIMA independently (e.g. for the borough-level actual-vs-forecast validation) can produce somewhat different results than the original run, particularly for volatile boroughs — a reminder that point forecasts from a single fit shouldn't be over-trusted.
- SARIMA vs Ensemble is not settled:** the two methods have each performed best in one of the two real validation windows tested so far (Naive/Ensemble in March–June, Ensemble alone in July). Aug–Dec forecasts from both methods are shown side by side rather than presenting either as definitively correct.
- External factors not included:** deprivation (IMD), weather, and monthly claimant count were all tested and show weak-to-moderate correlation with crime, but none meaningfully improved forecast accuracy when incorporated directly into the model.